

In this chapter we are going to set up the formalism to describe observables in quantum mechanics. This is an essential part of the formulation of the theory, as it deals with the description of the outcome of experiments. Beyond any theoretical sophistication, a physical theory is first and foremost a description of natural phenomena; therefore it requires a very precise framework that allows the observer to relate the outcome of experiments to theoretical predictions. As we will see, this is particularly true for quantum phenomena. The necessary formalism is very different from the intuitive one used in classical mechanics. A subtle point is that the state of the system is not an observable by itself. As seen in the [previous chapter](#), the state of the system is specified by a complex vector. Given a quantum system in a state that is described by a ket, we can perform measurements in that particular state; the results of these measurements are what we call observables. Therefore we need a formalism that allows us to determine:

- the possible outcomes of a measurement;
- what can be predicted about these outcomes, assuming that we know the complex vector that describes the state of the system.

In order to achieve these goals we will need to set up a new mathematical structure, based on operators that act on the state vectors. In this chapter you will learn the exact relation between states, operators and the outcome of experiments. This is a crucial step in building a framework that can yield predictions for the outcome of experiments, i.e. to connect the theory to the experiment. This chapter is focused on the general formalism and a few initial examples. More observables, like the angular momentum, will be introduced in detail in following chapters.

3.1 Introducing Observables

We define an observable O to be the outcome of an experiment that is measured by a given apparatus. The description of observables needs to be specified very precisely. This precise definition is encoded in a series of postulates that we are going to introduce in this chapter and discuss in detail. The overall picture can be summarised as follows.

For a quantum system in a state described by some complex vector $|\psi\rangle$, the outcome of a measurement of the observable O is a stochastic variable. The theory can predict:

1. the probability distribution function (pdf), p_O , that describes the possible results of the measurement and their respective frequencies;
2. the state of the system after the measurement.

We should note in passing that the knowledge of the pdf is equivalent to the knowledge of all its moments. Remember that the result of a measurement needs to be treated as a random variable and depending on the context we may focus either on its pdf or on its moments, i.e. we will want to compute its mean value, its variance, etc.

Mathematical Aside: Moments of Probability Distributions

Let us consider a random real variable r , characterised by a probability distribution function $p(r)$. The probability of finding r in the interval $I = [a, b]$ is

$$P(r \in I) = \int_a^b dx p(x). \quad (3.1)$$

(Note that x is a dummy variable of integration, we could use any name to identify the integration variable. If this does not sound familiar, think about it. ...)

The n th moment of the distribution p is defined as

$$\mu_n = \int dr p(r) r^n. \quad (3.2)$$

For a properly normalised probability distribution function:

$$\mu_0 = 1. \quad (3.3)$$

The mean value of the random variable r is

$$\mu_1 = \langle r \rangle = \int dr p(r) r. \quad (3.4)$$

The variance of the variable r is

$$\text{Var}[r] = \mu_2 - (\mu_1)^2. \quad (3.5)$$

Exercise 3.1.1 Compute the mean value and the variance of a Gaussian variable r , i.e. a random variable with probability distribution function

$$p(r) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x - x_0)^2}{2\sigma^2}\right]. \quad (3.6)$$

Note that the Gaussian distribution is completely determined by the knowledge of its first two moments.

3.1.1 A First Example: Position Operator

As discussed in the [previous chapter](#), the modulus squared of the wave function $|\psi(x)|^2$ describes the probability density of finding the system in the state $|\psi\rangle$ at x . So the outcome of a measurement of the position of the system, x , is a real stochastic variable and its pdf is $|\psi(x)|^2$. The mean value of the position of the system is therefore

$$\langle x \rangle = \int dx |\psi(x)|^2 x. \quad (3.7)$$

The crucial point here is the realisation that the mean value of the position of the system in the state $|\psi\rangle$, $\langle x \rangle$, can be written as the **expectation value** of a suitably defined operator $\hat{X}$ in the state $|\psi\rangle$, i.e. the matrix element of the operator $\hat{X}$ between $|\psi\rangle$ and $|\psi\rangle$.

The Hermitian operator $\hat{X}$ associated with the position of the system is such that the states $|x\rangle$ introduced in the [previous chapter](#) are eigenstates of $\hat{X}$ to the eigenvalue x – pay attention to the notation here! Using the fact that $|x\rangle$ are eigenstates of $\hat{X}$, we have

$$\langle x | \hat{X} | \psi \rangle = \left(\langle \psi | \hat{X} | x \rangle \right)^*, \quad (3.8)$$

$$= x \left(\langle \psi | x \rangle \right)^*, \quad (3.9)$$

$$= x \langle x | \psi \rangle; \quad (3.10)$$

with a slight abuse of notation, we can think of the operator $\hat{X}$ acting on the wave function as

$$\hat{X} \psi(x) = x \psi(x). \quad (3.11)$$

It is important to realise that [Eq. \(3.11\)](#) does not mean that $\psi(x)$ is an eigenfunction of $\hat{X}$ with eigenvalue x . The eigenvalue needs to be a constant that multiplies the state vector. The mean value in [Eq. \(3.7\)](#) can then be written as

$$\langle \hat{X} \rangle = \langle \psi | \hat{X} | \psi \rangle = \int dx \psi(x)^* x \psi(x) = \langle x \rangle. \quad (3.12)$$

As shown explicitly in the equation above, the expectation value in [Eq. \(3.12\)](#) depends on the state ψ . When there is no ambiguity, we shall omit the explicit ψ dependence, and write simply $\langle \hat{X} \rangle$, but remember that this is the mean value in a given state.

The mean value of the outcome of a measurement, $\langle x \rangle$, is therefore expressed as the expectation value of an operator acting in the space of physical states. We will now generalise this idea to a generic observable and discuss in detail the physical meaning of the mean value in the equation above.

3.1.2 Generic Observables

We define an **observable** to be a quantity that can be measured in an experiment. Experiments here are considered as black boxes: the system is prepared in some quantum state $|\psi\rangle$, then the experiment is performed and yields a value for the observable O . We will not dwell upon the details of the experimental settings, but we will assume that the experiment is reproducible, i.e. that it is possible to repeatedly prepare the system in

the same state and perform the same measurement. Following the example discussed in [Section 3.1.1](#) for the position of a quantum system, we are going to associate an operator with each observable. This is the **second postulate** of quantum mechanics.

Postulate 2 In quantum mechanics each observable O is associated with a *linear operator* $\hat{O}$ acting in the Hilbert space of physical states:

$$\begin{aligned}\hat{O}: \mathcal{H} &\rightarrow \mathcal{H} \\ |\psi\rangle &\mapsto |\psi'\rangle = \hat{O}|\psi\rangle.\end{aligned}\tag{3.13}$$

This is a weird statement! The postulate associates an operator with each observable. However, be aware that the measurement is NOT associated with acting with the operator on the state. By analogy with the example of the position operator, we expect that the mean value of the observable O in a quantum state can be computed as the expectation value of the operator $\hat{O}$ in that state:

$$\langle O \rangle = \langle \psi | \hat{O} | \psi \rangle.\tag{3.14}$$

The link between operators and the outcome of experiments needs to be explained in detail and this will be the focus of this chapter.

Example 3.1 The operator associated with the energy of the system is the Hamiltonian. The Hamiltonian of the system may be written as $\hat{H} = \hat{T} + \hat{V}$, where the kinetic and potential energy operators are defined by

$$\hat{T} = \frac{\hat{P}^2}{2m}, \quad \hat{V} = V(\hat{X}).\tag{3.15}$$

The operator $V(\hat{X})$, which is a function of the position operator $\hat{X}$, acts as

$$V(\hat{X})\psi(x) = V(x)\psi(x).\tag{3.16}$$

Any other operator that is a function of $\hat{X}$ can be defined in the same way. We shall see later how to define the operator $\hat{P}$ associated with the momentum of the system, and more complicated operators. $\hat{H}$ acts on state vectors and returns state vectors.

3.2 Observing Observables

Let us now explain in more detail the process of measurement in quantum mechanics and in particular what we mean when we say that the result of an experiment is a stochastic variable. If the observable O is measured several times under identical conditions, the result is a set of values $\{O^{(1)}, O^{(2)}, \dots, O^{(n)}\}$, where the suffix labels each measurement. This is a characteristic feature of quantum mechanics: the results of several independent measurements performed on a system prepared always in the same state $|\psi\rangle$ are different; they are distributed according to some pdf that can be predicted by the theoretical

framework. Hence the mean value of the observables in the given state can be computed by taking the average of the outcomes of the measurements, in the limit where the number of measurements becomes large. We see here the first explicit connection between the operator $\hat{O}$ and experimental results.

The expectation value of the operator $\hat{O}$ defined in Eq. (3.14) is equal to the average $\sum_{k=1}^n O^{(k)}/n$ in the limit where the number of measurements $n \rightarrow \infty$.

So far our examples have focused on the mean value of an observable, i.e. the first moment of its probability distribution. Moving beyond the first moment of the pdf, the theoretical framework of quantum mechanics allows us to predict:

1. the possible outcomes of the measurements above;
2. the probability of obtaining each of these possible outcomes.

It is important to realise that all questions in quantum mechanics need to be formulated in terms of possible outcomes of an experiment and their respective probabilities. These are the only quantities that are predicted by the theory. Trying to shortcut this procedure leads to well-known paradoxes.

Eigenvalues and Eigenstates

Our next step in building the theoretical framework is to specify the relation between the outcomes of measurements and some properties of the operator $\hat{O}$. This is encapsulated in the **third postulate** of quantum mechanics.

Postulate 3 The possible outcomes of experiments, O_k , are the **eigenvalues** of the operator $\hat{O}$, i.e. the solutions of the eigenvalue equation

$$\hat{O}|\psi_k\rangle = O_k|\psi_k\rangle, \quad (3.17)$$

where $|\psi_k\rangle$ is the **eigenstate** corresponding to the eigenvalue O_k . Note that an eigenstate is always associated with a specific eigenvalue.

Mathematical Aside: Eigenvalues and Degeneracies

An eigenvalue O_k of an operator $\hat{O}$ is called g -fold degenerate if there are exactly g linearly independent eigenvectors corresponding to the same eigenvalue:

$$\exists |u_k^{(n)}\rangle, \text{ such that } \hat{O}|u_k^{(n)}\rangle = O_k|u_k^{(n)}\rangle, \text{ for } n = 1, \dots, g. \quad (3.18)$$

Note that any linear combination

$$\sum_{n=1}^g c_n |u_k^{(n)}\rangle, \text{ with } c_n \in \mathbb{C}, \quad (3.19)$$

is also an eigenstate of $\hat{O}$ with the same eigenvalue O_k . The space of eigenstates corresponding to the eigenvalue O_k is a g -dimensional vector space.

The eigenfunction represents the state in which the measurement of O yields the value O_k with probability 1. To check this statement, we can compute the variance of O in the state $|\psi_k\rangle$:

$$\text{Var}_k[O] = \langle O^2 \rangle - \langle O \rangle^2, \quad (3.20)$$

$$\begin{aligned} &= \langle \psi_k | \hat{O}^2 | \psi_k \rangle - \langle \psi_k | \hat{O} | \psi_k \rangle^2, \\ &= O_k^2 \langle \psi_k | \psi_k \rangle - \left(O_k \langle \psi_k | \psi_k \rangle \right)^2 = 0, \end{aligned} \quad (3.21)$$

where we have used the fact that the eigenfunctions are normalised to one.

It is important in the equations above to distinguish the outcome of the i th measurement, denoted $O^{(i)}$, from the set of possible outcomes O_k . The latter are a property of the operator $\hat{O}$ and do not depend on the state of the system.

3.3 Hermitian Operators

We stated above that every observable is represented by an operator; ultimately we want to establish a correspondence

OBSERVABLE	$\longleftrightarrow$	OPERATOR
total energy	$\longleftrightarrow$	$\hat{H}$
position	$\longleftrightarrow$	$\hat{X}$
momentum	$\longleftrightarrow$	$\hat{P}$
$\vdots$	$\longleftrightarrow$	$\vdots$

Before we look into the details of associating observables with operators, we need to establish some properties of these operators. The main constraint comes from the fact that observables take **real** values only. Therefore we must require that the operators that represent observables have only **real eigenvalues**, since we want to identify the eigenvalues with the possible results of measurements. We can guarantee this if we only use **Hermitian** operators to represent observables.

3.3.1 Hermitian Conjugate

Let us define first the **Hermitian conjugate** $\hat{O}^\dagger$ of an operator $\hat{O}$.

Let $|\psi\rangle$ and $|\phi\rangle$ be arbitrary states in $\mathcal{H}$, then

$$\langle \phi | \hat{O}^\dagger | \psi \rangle = \left(\langle \psi | \hat{O} | \phi \rangle \right)^*. \quad (3.22)$$

In order to compute the matrix element of $\hat{O}^\dagger$ between $|\phi\rangle$ and $|\psi\rangle$, we need to compute the matrix element of $\hat{O}$ swapping the states and then taking the complex conjugate.

Mathematical Aside: A Comment on Hermitian Conjugation

Compare Eq. (3.22) with the more familiar expression from linear algebra:

$$O_{ij}^\dagger = O_{ji}^*. \quad (3.23)$$

The quantities in Eq. (3.22) are the matrix elements of the operator $\hat{O}$, just like O_{ij} are the matrix elements of a usual matrix O . The quantum states $|\phi\rangle$ and $|\psi\rangle$ are the ‘indices’ that label the matrix elements. Using this identification, many equations that we encounter in quantum mechanics become rather familiar.

Example 3.2 The operator $\hat{O} = \frac{d}{dx}$ is defined by specifying its action on the wave function of a quantum system:

$$\langle x|\hat{O}|\psi\rangle = \hat{O}\psi(x) = \frac{d}{dx}\psi(x). \quad (3.24)$$

Following our prescriptions, the matrix element of $\hat{O}$ between two states $|\psi\rangle$ and $|\phi\rangle$ is given by

$$\langle\psi|\hat{O}|\phi\rangle = \int dx \psi(x)^* \frac{d}{dx} \phi(x); \quad (3.25)$$

then we can integrate by parts to obtain

$$\int_{-\infty}^{\infty} \psi(x)^* \frac{d}{dx} \phi(x) dx = [\phi(x)\psi(x)^*]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \phi(x) \frac{d}{dx} \psi(x)^* dx. \quad (3.26)$$

We can discard the constant term on the right-hand side, since physically acceptable wave functions vanish at $x = \pm\infty$, and if we then take the complex conjugate of the resulting equation we obtain

$$\begin{aligned} \left(\int_{-\infty}^{\infty} \psi(x)^* \frac{d}{dx} \phi(x) dx \right)^* &= - \int_{-\infty}^{\infty} \phi(x)^* \frac{d}{dx} \psi(x) dx \\ &\equiv \int_{-\infty}^{\infty} \phi(x)^* \left(\frac{d}{dx} \right)^\dagger \psi(x) dx \end{aligned}$$

from the definition of Hermitian conjugate. Thus we can make the identification

$$\left(\frac{d}{dx} \right)^\dagger = -\frac{d}{dx}.$$

3.3.2 Hermitian Operators

We can now define a **Hermitian operator**. An operator $\hat{O}$ acting on a Hilbert space $\mathcal{H}$ is called **Hermitian** if

$$\hat{O}^\dagger \equiv \hat{O}. \quad (3.27)$$

Example 3.3 Equation (3.27) is clearly not true for all operators; $\frac{d}{dx}$ is NOT Hermitian since we have just shown that

$$\left(\frac{d}{dx}\right)^\dagger = -\frac{d}{dx},$$

whereas the operator $-i\hbar \frac{d}{dx}$ IS Hermitian; the proof is straightforward and is left as an exercise for the reader.

Strictly speaking, physical observables are associated with **self-adjoint** operators. In order to be able to define self-adjoint operators we first need to define the **domain**, $\mathcal{D}(\hat{O})$, of an operator $\hat{O}$. The domain $\mathcal{D}(\hat{O})$ is the subspace of $\mathcal{H}$ upon which the operator acts, i.e.

$$\hat{O}|\psi\rangle \in \mathcal{H} \implies |\psi\rangle \in \mathcal{D}(\hat{O}). \quad (3.28)$$

A self-adjoint operator is a Hermitian operator such that

$$\mathcal{D}(\hat{O}^\dagger) = \mathcal{D}(\hat{O}). \quad (3.29)$$

While the Hermitian operators that we encounter in quantum mechanics are mostly automatically self-adjoint, this pedantic distinction becomes relevant in some physically meaningful problems. Typically, problems with non-trivial boundary conditions require some care to extend a Hermitian operator so that it is actually self-adjoint.

3.4 Properties of Hermitian Operators

Hermitian operators obey properties that are important for building the logical framework of quantum mechanics.

1. Hermitian operators have *real* eigenvalues. The eigenvalue equation is

$$\hat{O}|\psi_k\rangle = O_k|\psi_k(x)\rangle, k = 1, \dots$$

$|\psi_k\rangle$ are the eigenfunctions of $\hat{O}$, O_k are the eigenvalues. Then we have

$$\hat{O} = \hat{O}^\dagger \implies O_k \in \mathbb{R}.$$

2. The eigenstates of a Hermitian operator that belong to different eigenvalues are orthogonal.
3. If $\hat{O}$ is a Hermitian operator acting on a vector space $\mathcal{H}$, there exists an orthogonal basis of $\mathcal{H}$ made up of eigenvectors of $\hat{O}$. In other words, every vector $|\psi\rangle$ can be expanded as

$$|\psi\rangle = \sum_k c_k |\psi_k\rangle, \quad (3.30)$$

where c_k are complex coefficients, computed by taking the projection of $|\psi\rangle$ onto the states $|\psi_k\rangle$:

$$c_k = \langle \psi_k | \psi \rangle.$$

As you can see from the equation above, for each state $|\psi\rangle$ there is a set of coefficients c_k ; they are the *coordinates* of the function $|\psi\rangle$ in the basis $\{|\psi_k\rangle, k = 1, \dots\}$. Do not confuse the coefficients c_k with the eigenvalues O_k ! The latter are a characteristic of the operator $\hat{O}$ and have nothing to do with the state ψ .

4. The commutator of two Hermitian operators is anti-Hermitian. While this property is trivial to prove, it is useful to keep in mind as a way to check your results. Every time you compute a commutator of Hermitian operators, you have a sanity check of your answer.

Mathematical Aside: Proof of the Properties

We shall now prove the first two properties above. The proofs are useful examples of manipulations involving operators acting on wave functions. Familiarity with these kinds of manipulations is essential for solving problems in quantum mechanics.

1. *Hermitian operators have real eigenvalues.*

Proof. Suppose $\hat{O}$ is a Hermitian operator so that $\hat{O}^\dagger = \hat{O}$, and let $\hat{O}$ have an eigenvalue O_k , with corresponding eigenfunction $|\psi_k\rangle$:

$$\hat{O} |\psi_k\rangle = O_k |\psi_k\rangle.$$

Then

$$\langle \psi_k | \hat{O} | \psi_k \rangle = O_k \langle \psi_k | \psi_k \rangle = O_k,$$

where in the last equality we have used the fact that the ket $|\psi_k\rangle$ is properly normalised to one.

If we take the complex conjugate of this equation, we obtain

$$\left(\langle \psi_k | \hat{O} | \psi_k \rangle \right)^* = O_k^*,$$

but if we make use of the definition of the Hermitian conjugate, we can rewrite the left-hand side of this equation in terms of $\hat{O}^\dagger$ and use the fact that $\hat{O}^\dagger = \hat{O}$ by hypothesis:

$$\left(\langle \psi_k | \hat{O} | \psi_k \rangle \right)^* = \langle \psi_k | \hat{O}^\dagger | \psi_k \rangle = \langle \psi_k | \hat{O} | \psi_k \rangle.$$

The right-hand side is now just the matrix element that appears in the first equation and is equal to O_k , so we have proved that

$$O_k^* = O_k,$$

thus showing that the eigenvalue O_k is real as stated.

2. *The eigenfunctions of a Hermitian operator which belong to different eigenvalues are orthogonal.*

Proof. Suppose that

$$\hat{O} |\psi_1\rangle = O_1 |\psi_1\rangle \quad \text{and} \quad (3.31)$$

$$\hat{O} |\psi_2\rangle = O_2 |\psi_2\rangle \quad \text{with } O_1 \neq O_2. \quad (3.32)$$

From Eq. (3.31) we have

$$\langle \psi_2 | \hat{O} | \psi_1 \rangle = O_1 \langle \psi_2 | \psi_1 \rangle, \quad (3.33)$$

whereas from Eq. (3.32)

$$\langle \psi_1 | \hat{O} | \psi_2 \rangle = O_2 \langle \psi_1 | \psi_2 \rangle. \quad (3.34)$$

Taking the complex conjugate of Eq. (3.34) yields on the left-hand side

$$\left(\langle \psi_1 | \hat{O} | \psi_2 \rangle \right)^* \equiv \langle \psi_2 | \hat{O}^\dagger | \psi_1 \rangle = \langle \psi_2 | \hat{O} | \psi_1 \rangle, \quad (3.35)$$

whereas the right-hand side gives

$$O_2^* \langle \psi_2 | \psi_1 \rangle = O_2 \langle \psi_2 | \psi_1 \rangle, \quad (3.36)$$

using the fact that $O_2 = O_2^*$.

Comparing with Eq. (3.33) we see that

$$O_2 \langle \psi_2 | \psi_1 \rangle = O_1 \langle \psi_2 | \psi_1 \rangle, \quad (3.37)$$

which we can rearrange to yield the result

$$(O_2 - O_1) \langle \psi_2 | \psi_1 \rangle = 0. \quad (3.38)$$

Given that $O_2 \neq O_1$ by hypothesis, this implies that

$$\langle \psi_2 | \psi_1 \rangle = 0, \quad (3.39)$$

which is the desired result.

3.5 Spectral Decomposition

Using the fact that we can define a basis in the space of physical states made up of eigenstates of the operator $\hat{O}$ as shown in Eq. (3.30), we can formulate the rule to compute the probability of finding any given eigenvalue when performing a measurement. This is the **fourth postulate** of quantum mechanics.

Postulate 4 Given the decomposition in Eq. (3.30), the probability of finding the nondegenerate eigenvalue O_k when measuring O in the state described by the normalised vector $|\psi\rangle$ is given by

$$P_k = |c_k|^2. \quad (3.40)$$

Clearly the sum of probabilities should be properly normalised and therefore

$$\sum_k P_k = \sum_k |c_k|^2 = 1. \quad (3.41)$$

This is simply a confirmation of the fact that the state vector $|\psi\rangle$ is normalised to one.

Example 3.4 Note that the concept of superposition of states is very different from anything we have encountered in classical mechanics. Consider two quantum states $|A\rangle$ and $|B\rangle$, such that the measurement of an observable O yields the result a with probability 1 when the system is in the state $|A\rangle$, and the result b with probability 1 when the system is in the state $|B\rangle$. The superposition principle states that the state vector

$$|C\rangle = c_A|A\rangle + c_B|B\rangle, \quad (3.42)$$

where c_A and c_B are complex numbers such that $|c_A|^2 + |c_B|^2 \neq 0$, describes a possible physical state of the system. According to the postulates that we formulated so far, the measurement of the observable O in the state $|C\rangle$ can only yield the value a or b , with respective probabilities

$$P_a = \frac{|c_A|^2}{|c_A|^2 + |c_B|^2}, \quad P_b = \frac{|c_B|^2}{|c_A|^2 + |c_B|^2}. \quad (3.43)$$

No other results are possible for the measured value of O in the state $|C\rangle$.

Example 3.5 Let us consider again the discretised one-dimensional system in Fig. 2.3; we can have a state $|1\rangle$ where the particle is localised for example at site 1, and a state $|2\rangle$ where the particle is localised at site 2. Measuring the position of the particle in state $|1\rangle$ yields $x = 1$ with probability 1. Likewise we obtain $x = 2$ with probability 1 for a particle described by the state vector $|2\rangle$.

The state $\frac{1}{\sqrt{2}}|1\rangle + \frac{1}{\sqrt{2}}|2\rangle$ is an admissible quantum state. Measuring the position of the particle in this latter state, the outcome will be $x = 1$ or $x = 2$ with 50% probability. No other value is allowed in this state.

Degenerate Eigenvalues

In the case where the eigenvalue O_k is g -fold degenerate, the probability of O_k being the result of a measurement of the observable O in the state $|\psi\rangle$ is obtained by summing over the contributions from the whole subspace spanned by the degenerate eigenvectors. Following the notation introduced in Eq. (3.18), the probability is

$$P_k = \sum_{n=1}^g \left| \langle u_k^{(n)} | \psi \rangle \right|^2. \quad (3.44)$$

3.6 Collapse of the State Vector

Given the state vector $|\psi\rangle$, the fourth postulate of quantum mechanics allows us to compute the probability of any given outcome for the measurement of an observable. However, after the measurement has been performed, the outcome of the experiment is known with probability 1, and therefore we must conclude that the state of the system has changed. This is an important aspect of measurement in quantum mechanics: the measurement must change the state of the system. This is summarised in the **fifth postulate** of quantum mechanics.

Postulate 5 Immediately after a measurement that gave the result O_k , where O_k is a nondegenerate eigenvalue, the system is in the state $|\psi_k\rangle$, the eigenvector of $\hat{O}$ associated with the eigenvalue O_k . The state vector has been *projected* onto the eigenstate by the process of performing the measurement.

We will discuss more precisely what we mean by *immediately after* when we discuss the time evolution of quantum systems in the [next chapter](#).

We can express the same concept using the projection operator introduced in [Eq. \(2.66\)](#); immediately after a measurement yielding the value O_k , the state of the system is transformed according to

$$|\psi\rangle \mapsto \frac{1}{\sqrt{\langle\psi|\hat{P}_k|\psi\rangle}}\hat{P}_k|\psi\rangle, \quad (3.45)$$

where we have explicitly normalised the projected vector to have unit norm. This is sometimes referred to as the *collapse of the state vector*. Clearly this is an idealised description of a much more complicated process where a ‘classical’ instrument and the quantum system interact. While a fully satisfactory description of quantum measurement is a subtle issue, the fifth postulate gives a practical recipe, which underlies the numerous successful predictions of quantum mechanics.

After a measurement that yielded the value O_k , the wave function of the system coincides with the eigenfunction $|\psi_k\rangle$. Then, as discussed below [Eq. \(3.17\)](#), if we perform immediately another measurement of O we will find the *same* value O_k with probability 1.

Conversely, if the wave function does not coincide with one of the eigenfunctions, then the observable O does not have a given value in the state $|\psi\rangle$. We can only compute the probability for each eigenvalue to be the outcome of the experiment.

Clearly these phenomena do not have a classical analogue. The description of a physical system in quantum mechanics is radically different from the classical one. You need to practice in order to get familiar with the quantum-mechanical framework.

Degenerate Eigenvalues

In the case of degenerate eigenvalues, the state vector before the measurement can be expanded as

$$|\psi\rangle = \sum_k \sum_{n=1}^{g_k} c_{kn} |u_k^{(n)}\rangle, \quad (3.46)$$

where the index k runs over all eigenvalues, and for each eigenvalue we have a sum over n running over the number of degenerate eigenvectors that correspond to the k th eigenvalue.

After the measurement the state of the system is still given by Eq. (3.45), but the projector $\hat{\mathcal{P}}_k$ needs to project on the subspace of degenerate eigenstates, i.e.

$$\hat{\mathcal{P}}_k = \sum_{n=1}^{g_k} |u_k^{(n)}\rangle \langle u_k^{(n)}|. \quad (3.47)$$

Exercise 3.6.1 Show that the projected (and normalised) state according to Eq. (3.47) is

$$\frac{1}{\sqrt{\sum_{n=1}^{g_k} |c_{kn}|^2}} \sum_{n=1}^{g_k} c_{kn} |u_k^{(n)}\rangle. \quad (3.48)$$

3.7 Compatible Observables

Suppose A and B are observables and we perform the following sequence of measurements in succession on a single system:

1. measure A
2. measure B
3. remeasure A

Then *if and only if* the result of 3 is *certain* to be the same as the result of 1, we say that A and B are **compatible observables**.

In general, this will not be the case: according to our previous discussion about the collapse of the state vector, the measurement of B will project the state of the system onto an eigenstate of B , and therefore ‘spoil’ the result of 1. Let us analyse this statement in a little more detail, following the ideas that we have introduced so far about observables and measurements. Suppose that A and B are represented by operators $\hat{A}$ and $\hat{B}$, respectively, with eigensystems

$$\begin{aligned} \hat{A} |u_i\rangle &= A_i |u_i\rangle, \\ \hat{B} |v_j\rangle &= B_j |v_j\rangle. \end{aligned}$$

For simplicity we are going to consider the case of nondegenerate eigenvalues – taking into account degeneracies adds a little complexity for no intellectual gain. Measurement 1 must return one of the eigenvalues of the operator $\hat{A}$, A_j say, collapsing the system into the state $|u_j\rangle$. Measurement 2 yields an eigenvalue of $\hat{B}$, B_k , forcing the system into the state $|v_k\rangle$, so that measurement 3 is made with the system in the state $|v_k\rangle$. The only way that 3 is *certain* to yield the result A_j as obtained in 1 is if $|v_k\rangle \equiv |u_j\rangle$. For this to be true in all circumstances it must be the case that each eigenvector $|v_k\rangle$ of $\hat{B}$ is identical to some eigenvector $|u_j\rangle$ of $\hat{A}$. If there is no degeneracy this implies a one-to-one correspondence between the eigenvectors of $\hat{A}$ and the eigenvectors of $\hat{B}$. We say that $\hat{A}$ and $\hat{B}$ have a *common eigenbasis*. These properties are summarised in the so-called **compatibility theorem**.

Theorem 3.1 *Given two observables, A and B , represented by Hermitian operators $\hat{A}$ and $\hat{B}$, then any one of the following three statements implies the other two:*

1. A and B are compatible observables;
2. $\hat{A}$ and $\hat{B}$ have a common eigenbasis;
3. the operators $\hat{A}$ and $\hat{B}$ commute: $[\hat{A}, \hat{B}] = 0$.

Example Proof. Let us show, for instance, that $3 \Rightarrow 2$. We have

$$\begin{aligned}\hat{A} |u_i\rangle &= A_i |u_i\rangle, \\ \hat{B} |v_i\rangle &= B_i |v_i\rangle,\end{aligned}$$

so that for *any* eigenvector of $\hat{A}$

$$\begin{aligned}\hat{A}\hat{B} |u_i\rangle &= \hat{B}\hat{A} |u_i\rangle \quad \text{by virtue of 3} \\ &= \hat{B} A_i |u_i\rangle \\ &= A_i \hat{B} |u_i\rangle.\end{aligned}$$

Thus $\hat{B} |u_i\rangle$ is an eigenvector of $\hat{A}$ belonging to the eigenvalue A_i . If we assume that the eigenvalues are nondegenerate, then $\hat{B} |u_i\rangle$ must be some multiple of $|u_i\rangle$:

$$\hat{B} |u_i\rangle = \rho |u_i\rangle, \quad \text{say.} \quad (3.49)$$

This just says that $|u_i\rangle$ is an eigenstate of $\hat{B}$ belonging to the eigenvalue ρ , and we must have that, for some j ,

$$\rho = B_j \quad \text{and} \quad |u_i\rangle = |v_j\rangle. \quad (3.50)$$

Thus any eigenstate of the set $\{|u_i\rangle\}$ coincides with some member of the set $\{|v_j\rangle\}$. The correspondence has to be one-to-one because both sets are orthonormal; if we assume that two states in one set coincide with a single state in the other set, we are led to a contradiction that two orthogonal vectors are identical to the same vector. By simply relabelling all the vectors in one set we can always ensure that

$$|u_1\rangle = |v_1\rangle, |u_2\rangle = |v_2\rangle, |u_3\rangle = |v_3\rangle, \dots \text{etc.}, \quad (3.51)$$

and this is the common eigenbasis. A more general proof, in the case where the eigenvalues are degenerate, is left as an exercise.

3.8 Complete Sets of Commuting Observables

Consider an observable A , and a basis made up of eigenstates of $\hat{A}$, $\{|u_1\rangle, |u_2\rangle, \dots\}$. If all the eigenvalues are nondegenerate, each eigenvalue identifies uniquely *one* eigenstate. Hence we can label the eigenstates by their eigenvalue; if

$$\hat{A}|u_n\rangle = a_n|u_n\rangle, \quad (3.52)$$

then we can rename

$$|u_n\rangle \equiv |a_n\rangle. \quad (3.53)$$

In this case, the observable A constitutes by itself a **complete set of commuting observables** (CSCO), i.e. the eigenvalues of $\hat{A}$ are sufficient to identify the eigenvectors that form a basis of the space of physical states.

However, this is no longer true if some of the eigenvalues are degenerate, since in this case there are several eigenstates corresponding to the same degenerate eigenvalue. In order to distinguish these eigenstates, we can use the eigenvalues of a second observable B , which commutes with A . According to the compatibility theorem, we can find a basis of common eigenstates of $\hat{A}$ and $\hat{B}$. If each pair of eigenvalues $\{a_n, b_p\}$ identifies uniquely *one* vector of the basis, then the set $\{A, B\}$ is a CSCO. If this is not the case, then there must be at least one pair $\{a_n, b_p\}$ for which there exists more than one eigenvector with these eigenvalues, i.e. there exist at least two vectors $|w_1\rangle$ and $|w_2\rangle$, such that

$$\hat{A}|w_1\rangle = a_n|w_1\rangle, \quad \hat{B}|w_1\rangle = b_p|w_1\rangle, \quad (3.54)$$

$$\hat{A}|w_2\rangle = a_n|w_2\rangle, \quad \hat{B}|w_2\rangle = b_p|w_2\rangle. \quad (3.55)$$

In this case specifying the values of a_n and b_p is not sufficient to identify uniquely one eigenvector, since any linear combination of $|w_1\rangle$ and $|w_2\rangle$ is also a simultaneous eigenvector of $\hat{A}$ and $\hat{B}$ with the same eigenvalues. Iterating the above procedure, we add to our set of observables one more quantity C , which commutes with both A and B , and we choose a basis made up of simultaneous eigenvalues of the three operators $\hat{A}, \hat{B}, \hat{C}$. If each eigenstate in the basis is uniquely identified by the set of eigenvalues $\{a_n, b_p, c_q\}$, then $\{A, B, C\}$ is a CSCO. If not, we need to add one more observable to our set, and so on.

A set of observables $A, B, C, \dots$ is called a CSCO if:

- i. all the observables commute by pairs;
- ii. specifying the eigenvalues of all the operators in the CSCO identifies a unique common eigenvector.

Given a CSCO, we can choose a basis for the space of states made up of common eigenvectors of the operators associated with the observables. Each eigenvector is uniquely identified by the values of the eigenvalues to which it corresponds:

$$\begin{aligned} \hat{A}|a_n, b_p, c_q, \dots\rangle &= a_n|a_n, b_p, c_q, \dots\rangle, \\ \hat{B}|a_n, b_p, c_q, \dots\rangle &= b_p|a_n, b_p, c_q, \dots\rangle, \\ \hat{C}|a_n, b_p, c_q, \dots\rangle &= c_q|a_n, b_p, c_q, \dots\rangle, \\ &\dots \end{aligned} \quad (3.56)$$

Given a CSCO, we can expand any generic wave function in the basis of common eigenstates labelled by the eigenvalues of the observables:

$$|\psi\rangle = \sum_{n,p,q} \psi_{n,p,q} |a_n, b_p, c_q\rangle.$$

The modulus square of the coefficients, $|\psi_{n,p,q}|^2$, yields the probability of finding simultaneously the values a_n, b_p, c_q if we measure A, B, C in the state $|\psi\rangle$.

3.9 Continuous Spectrum

Until now we have discussed a number of examples where operators have a discrete spectrum, i.e. where the eigenvalues are numbered by some integer index k .¹ However, there are operators that have a continuous spectrum, like the energy and the momentum of an unbound state, or the position operator $\hat{X}$. In order to deal with these cases, we need to generalise the formalism that we have introduced so far. As you will see below, the modifications are minimal, and rather straightforward.

3.9.1 Eigenvalue Equation

Let us denote by $\hat{f}$ the Hermitian operator associated with an observable with a continuous spectrum; the eigenvalue equation takes the form

$$\hat{f}|f\rangle = f|f\rangle. \quad (3.57)$$

Note that in Eq. (3.57) we have denoted the eigenstate simply by $|f\rangle$. The set of eigenvectors is a complete set, and we can write the completeness relation by substituting the sum over the eigenvalues with an integral, viz.

$$\int df |f\rangle\langle f| = 1. \quad (3.58)$$

Once again, using the completeness relation, a generic state can be expanded as a superposition of eigenstates:

$$|\psi\rangle = \int df \psi(f)|f\rangle; \quad (3.59)$$

you should compare this expression with its analogue Eq. (3.30) in the case of discrete eigenvalues. Here we are implicitly assuming that the integral over vectors in $\mathcal{H}$ can be properly defined and yields another element of the same vector space $\mathcal{H}$. The coefficients in the expansion Eq. (3.59) are obtained by taking the scalar product

$$\psi(f) = \langle f|\psi\rangle. \quad (3.60)$$

Example 3.6 For a quantum system on a line, we can use the completeness relation

$$\int dx |x\rangle\langle x| = 1 \quad (3.61)$$

¹ In the case of a discrete spectrum, the total number of eigenvalues may well be infinite, however, the eigenvalues are labelled by integer numbers.

and rewrite Eq. (3.60) as

$$\langle f|\psi\rangle = \int dx \langle f|x\rangle \langle x|\psi\rangle = \int dx f(x)^* \psi(x). \quad (3.62)$$

The probabilistic interpretation of the state vector, i.e. the fourth postulate of quantum mechanics, can be generalised to the case of a continuous spectrum. The probability of finding a result between f and $f + df$ when measuring the observable f is given by

$$|\psi(f)|^2 df.$$

Thus we derive the normalisation condition

$$\int df |\psi(f)|^2 = 1. \quad (3.63)$$

The integral in Eq. (3.59) defines a normalisable state if and only if the function $\psi(f)$ is square-integrable.

3.9.2 Orthonormality

Following the steps performed to obtain Eq. (3.39), we have

$$\langle f'|\hat{f}|f\rangle = f\langle f'|f\rangle. \quad (3.64)$$

Taking the complex conjugate of Eq. (3.64), and using the fact that $\hat{f}$ is Hermitian, yields

$$(\langle f|\hat{f}|f'\rangle)^* = \langle f'|\hat{f}^\dagger|f\rangle, \quad (3.65)$$

$$= \langle f'|\hat{f}|f\rangle. \quad (3.66)$$

Combining the results above:

$$(f - f') \langle f'|f\rangle = 0, \quad (3.67)$$

and therefore, if $f \neq f'$,

$$\langle f'|f\rangle = 0. \quad (3.68)$$

Now comes a subtle point. The norm of the state $|\psi\rangle$ is given by

$$\langle \psi|\psi\rangle = \int df df' \psi(f)^* \psi(f') \langle f|f'\rangle, \quad (3.69)$$

$$= \int df |\psi(f)|^2 \int df' \langle f|f'\rangle, \quad (3.70)$$

where we used the orthogonality result Eq. (3.68). Now the integral over df' in Eq. (3.70) vanishes for any finite value of $\langle f|f\rangle$. Therefore we need to impose that $\langle f|f\rangle$ is infinite and normalised so that

$$\int df' \langle f'|f\rangle = 1. \quad (3.71)$$

The **delta function** introduced by Dirac satisfies precisely this condition, and we can require

$$\langle f' | f \rangle = \delta(f - f'). \quad (3.72)$$

The eigenstates of the continuous spectrum have infinite norm, and therefore cannot be considered as genuine physical states. However, they provide a useful basis for expanding normalisable state vectors that correspond to physical states, as in [Eq. \(3.59\)](#).

3.9.3 Spectral Decomposition

For a generic operator $\hat{O}$, the eigenvalue spectrum can be made up of discrete eigenvalues O_n and continuous values f . In this case we need to consider both the eigenfunctions of the discrete spectrum and those of the continuous spectrum to have a basis to expand quantum states in:

$$|\psi\rangle = \sum_n \langle n | \psi \rangle |n\rangle + \int df \langle f | \psi \rangle |f\rangle. \quad (3.73)$$

[Equation \(3.73\)](#) summarises the completeness relation for both discrete and continuous spectra.

Taking the scalar product with eigenstates of the position eigenstates $|x\rangle$:

$$\psi(x) = \sum_n \psi_n u_n(x) + \int df \psi(f) u_f(x), \quad (3.74)$$

where

$$\psi_n = \langle n | \psi \rangle, \quad u_n(x) = \langle x | n \rangle, \quad (3.75)$$

$$\psi(f) = \langle f | \psi \rangle, \quad u_f(x) = \langle x | f \rangle. \quad (3.76)$$

Example 3.7 Gaussian wave functions. For a quantum system on a line, the position operator $\hat{X}$ has a continuous spectrum. The eigenvalue equation

$$\hat{X} |x\rangle = x |x\rangle \quad (3.77)$$

has a solution for each value of x . The eigenvectors are normalised to the Dirac delta:

$$\langle x | x' \rangle = \delta(x - x'), \quad (3.78)$$

and a generic vector can be expanded as

$$|\psi\rangle = \int dx |x\rangle \langle x | \psi \rangle = \int dx \psi(x) |x\rangle. \quad (3.79)$$

Consider now the state defined by the wave function

$$\psi(x) = \langle x | \psi \rangle = C \exp \left[\frac{i}{\hbar} p_0 x - \frac{(x - x_0)^2}{2\xi^2} \right]. \quad (3.80)$$

The state has a finite norm:

$$\int dx |\psi(x)|^2 = |C|^2 \sqrt{\pi} \xi, \quad (3.81)$$

which yields the normalisation constant

$$C = \frac{1}{\pi^{1/4} \sqrt{\xi}}. \quad (3.82)$$

Having properly normalised the state vector, we can compute the mean value of x , i.e. the average of multiple measurements of the position of the system prepared in the state $|\psi\rangle$, specified by Eq. (3.80):

$$\langle x \rangle = \int dx |\psi(x)|^2 x = x_0. \quad (3.83)$$

The variance of the same ensemble of measurements is

$$\langle \Delta x^2 \rangle = \langle (x - x_0)^2 \rangle = \int dx |\psi(x)|^2 (x - x_0)^2 = \xi^2. \quad (3.84)$$

In the last two equations, the quantities on the left-hand side are obtained by preparing the system several times in the same state $|\psi\rangle$, performing measurements and taking averages. The quantities on the right-hand side, i.e. x_0 and ξ , are parameters that define the state of the system. By performing measurements, and looking at the moments of the observables, we can determine these parameters, i.e. we learn about the state of the system.

3.10 Momentum Operator

The *momentum operator* is defined as a differential operator acting on the wave function:

$$\hat{P}\psi(x) = \langle x | \hat{P} | \psi \rangle = -i\hbar \frac{d}{dx} \psi(x). \quad (3.85)$$

This can be seen as a realisation of de Broglie's duality hypothesis; according to wave-particle duality, with a particle with momentum p we can associate a wave with wavelength h/p . A wave with a fixed wavelength is a plane wave, described by the function

$$\psi_p(x) = C \exp[ipx/\hbar]. \quad (3.86)$$

When we act with the operator $\hat{P}$ defined in Eq. (3.85), we see that $\psi_p(x)$ is an eigenstate of $\hat{P}$ with eigenvalue p , and therefore we can associate the plane wave with a state with given momentum p .

Note that there are no restrictions on the possible values of p . The spectrum of $\hat{P}$ is therefore a continuous spectrum. The normalisation of the eigenstates is

$$\begin{aligned}\langle p'|p\rangle &= \int dx \langle p'|x\rangle \langle x|p\rangle \\ &= |C|^2 \int dx \exp \left[\frac{i}{\hbar} (p - p')x \right] \\ &= |C|^2 2\pi\hbar \delta(p - p').\end{aligned}\tag{3.87}$$

Following the convention we set in [Section 3.9.2](#), we can choose $C = 1/\sqrt{2\pi\hbar}$, and hence

$$\psi_p(x) = \langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} \exp[ipx/\hbar].\tag{3.88}$$

Example 3.8 We have seen previously that the action of the position operator $\hat{X}$ is

$$\hat{X}\psi(x) = x\psi(x),\tag{3.89}$$

i.e. the wave function is simply multiplied by the value of x . Consider the case $\hat{O}_1 = \hat{X}$, $\hat{O}_2 = \frac{d}{dx}$. Then

$$\hat{O}_1\hat{O}_2\psi(x) = \hat{X}\left(\frac{d}{dx}\psi(x)\right),\tag{3.90}$$

$$= x\frac{d}{dx}\psi(x),\tag{3.91}$$

while

$$\hat{O}_2\hat{O}_1\psi(x) = \hat{O}_2(\hat{X}\psi(x)),\tag{3.92}$$

$$= \hat{O}_2(x\psi(x)),\tag{3.93}$$

$$= \frac{d}{dx}(x\psi(x)),\tag{3.94}$$

$$= \psi(x) + x\frac{d}{dx}\psi(x).\tag{3.95}$$

Putting the two results together, we obtain for this particular choice of $\hat{O}_1$ and $\hat{O}_2$:

$$[\hat{O}_1, \hat{O}_2]\psi(x) = -\psi(x),\tag{3.96}$$

i.e.

$$[\hat{O}_1, \hat{O}_2] = -1.\tag{3.97}$$

From the example above we deduce the fundamental **canonical commutation relation**:

$$[\hat{X}, \hat{P}] = i\hbar.\tag{3.98}$$

3.10.1 Momentum as Generator of Translations

Consider the wave function of a one-dimensional system, and translate the coordinate x by some amount a . Expanding at first order in a :

$$\psi(x+a) = \psi(x) + a \frac{d}{dx} \psi(x) + O(a^2). \quad (3.99)$$

We want to describe the change in the wave function by the action of a Hermitian operator acting on it. As discussed above, the differential operator d/dx is anti-Hermitian; we can introduce a Hermitian operator by rewriting Eq. (3.99) as

$$\begin{aligned} \delta_a \psi(x) &= \psi(x+a) - \psi(x) \\ &= \frac{i}{\hbar} a \left(-i\hbar \frac{d}{dx} \right) \psi(x) + O(a^2) \\ &= ia \left(\frac{\hat{p}}{\hbar} \right) \psi(x) + O(a^2). \end{aligned} \quad (3.100)$$

The quantity in the bracket on the right-hand side of Eq. 3.100 is a Hermitian operator. It can easily be checked that it has dimensions of inverse length, and actually coincides with the momentum operator that we defined above in order to implement de Broglie's idea of duality.

If we only consider infinitesimal transformations and neglect the non-linear dependence on the transformation parameter, we can write

$$\delta_a \psi(x) = ia \hat{T} \psi(x). \quad (3.101)$$

The operator $\hat{T}$ is called the **generator** of translations. The momentum operator coincides with the generator of translations up to the factor $1/\hbar$, which takes care of dimensions.

3.10.2 Position and Momentum Representations

So far we have adopted a heuristic approach in order to define the position and momentum operators, based on physical considerations. It is worthwhile having a more formal introduction, which follows closely the ideas of vectors and bases that we have developed in this book.

The state of the quantum system is described by a vector $|\psi\rangle$. This vector can be expanded in a basis made up of eigenstates of the position operator $\{|x\rangle\}$:

$$|\psi\rangle = \int dx |x\rangle \langle x|\psi\rangle, \quad (3.102)$$

and the vector is fully specified by specifying its coordinates in that basis:

$$\psi(x) = \langle x|\psi\rangle. \quad (3.103)$$

This is called the **position representation** of the state.

The same vector can be expanded in eigenstates of momentum:

$$|\psi\rangle = \int dp |p\rangle \langle p|\psi\rangle. \quad (3.104)$$

In this case the vector is fully described by the wave function in the so-called **momentum representation**:

$$\tilde{\psi}(p) = \langle p | \psi \rangle. \quad (3.105)$$

Using the completeness relations allows us to write

$$\tilde{\psi}(p) = \int dx \langle p | x \rangle \langle x | \psi \rangle, \quad (3.106)$$

$$= \int \frac{dx}{\sqrt{2\pi\hbar}} e^{-i\frac{px}{\hbar}} \psi(x), \quad (3.107)$$

which shows that the two representations are related by Fourier transforms.

The vector $|\psi\rangle$ has finite norm, while the eigenstates of both position and momentum are states with an infinite norm and do not represent physical states. Nonetheless, they are useful as a basis to expand the state vector and define a pdf for observing given values of x and p in the state $|\psi\rangle$.

3.10.3 Wave Packets

Physical states, i.e. state vectors with a finite norm, can be constructed as linear superpositions of momentum eigenstates. These states are called **wave packets**:

$$|W\rangle = \int dp g(p) |p\rangle, \quad g(p) \in \mathbb{C}. \quad (3.108)$$

For the wave packet to be normalisable, we require

$$\begin{aligned} \langle W | W \rangle &= \int dp' dp g(p')^* g(p) \langle p' | p \rangle \\ &= \int dp |g(p)|^2 < \infty. \end{aligned} \quad (3.109)$$

If we choose the function $g(p)$ such that $|g(p)|^2$ is peaked around some value p_0 , then the probability of measuring a given value for the momentum of the system in the state $|W\rangle$ has a maximum at p_0 and rapidly goes to zero as we move away from p_0 . While the wave packet is not an eigenstate of momentum, we can think of it as a physical realisation of a state with momentum approximately equal to p_0 . Making the distribution more peaked around its maximum yields a more ‘collimated’ beam.

In this section we are going to focus on Gaussian wave packets, normalised to one, viz.

$$g(p) = \frac{1}{(2\pi\sigma^2)^{1/4}} \exp \left[-\frac{1}{4\sigma^2} (p - p_0)^2 \right]. \quad (3.110)$$

Exercise 3.10.1 Check that the wave packet is properly normalised, $\langle W|W \rangle = 1$.
Compute the expectation values

$$\begin{aligned}\langle W|\hat{P}|W \rangle &= \int dp' dp g(p')^* g(p) \langle p'|\hat{P}|p \rangle \\ &= \int dp |g(p)|^2 p = p_0, \\ \langle W|\hat{P}^2|W \rangle &= p_0^2 + \sigma^2.\end{aligned}$$

Using the results of the exercise above, we can determine the uncertainty on the momentum

$$\begin{aligned}\Delta p &= \sqrt{\langle W|\hat{P}^2|W \rangle - \langle W|\hat{P}|W \rangle^2} \\ &= \sigma.\end{aligned}\tag{3.111}$$

Starting from Eq. (3.108), we can readily compute

$$\langle p|W \rangle = \int dp' g(p') \langle p|p' \rangle = g(p),\tag{3.112}$$

which is sometimes referred to as the wave function in momentum space.

A slightly lengthier computation yields the wave function in position space:

$$\langle x|W \rangle = \int dp \frac{1}{(2\pi\sigma^2)^{1/4}} \exp\left[-\frac{1}{4\sigma^2}(p-p_0)^2\right] \frac{1}{(2\pi\hbar)^{1/2}} \exp\left[i\frac{px}{\hbar}\right],\tag{3.113}$$

$$= \left(\frac{2\sigma^2}{\pi\hbar^2}\right)^{1/4} \exp\left[-\frac{\sigma^2 x^2}{\hbar^2} + i\frac{p_0 x}{\hbar}\right].\tag{3.114}$$

Exercise 3.10.2 Compute the integral yielding the wave function in Eq. (3.114). Check that the modulus square of the wave function is properly normalised to one, as required by the Parseval identity.

Using the explicit expression for the wave function in position space yields

$$\begin{aligned}\langle W|\hat{X}|W \rangle &= 0, \\ \langle W|\hat{X}^2|W \rangle &= \frac{\hbar^2}{4\sigma^2},\end{aligned}\tag{3.115}$$

and therefore the uncertainty on the position of the system in the state $|W \rangle$:

$$\Delta x = \frac{\hbar}{2\sigma}.\tag{3.116}$$

3.11 The Uncertainty Principle

In classical mechanics the state of a particle in a one-dimensional world is completely determined by the value of its position $x(t)$ and momentum $p(t)$, i.e. by its *trajectory*.

The situation is radically different in quantum mechanics. The probabilistic interpretation of the wave function implies that we can at best obtain the probability density for a particle to be at a given position x at time t . As a consequence, the concept of *classical trajectory* used in Newtonian mechanics does not make sense in quantum mechanics. The position and momentum of the particle can be defined, but their values cannot be measured simultaneously with arbitrary precision. As expected, when the scales in the problem are much larger than the Planck constant $\hbar$, the classical results are recovered.

These two features are summarised in the so-called **uncertainty relations**, first derived by Heisenberg. The uncertainty relations state that the standard deviations Δx and Δp of the measured position and momentum in a given quantum state satisfy

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}. \quad (3.117)$$

The derivation of this inequality is left as an exercise for the reader, see Problem 3.12 below. It is clear from Eq. (3.117) that if the position of the particle is known exactly, i.e. if the state is an eigenstate of position, then the knowledge of its momentum is completely lost, in the sense that the standard deviation of the measured momentum is divergent. In general, the product of the two uncertainties has to be greater than $\hbar/2$. Note that the Gaussian wave packet discussed in Section 3.10.3 saturates the bounds exactly.

It is important to appreciate that Heisenberg's inequalities reflect a physical limitation. The outcomes of measurements are stochastic variables, the state of the quantum system determines the expectation value and the standard deviation of these stochastic variables. A better experimental apparatus would **not** allow a higher precision to be obtained. The uncertainty is a quantum-mechanical property of the system.

The uncertainty principle also encodes the idea that in quantum mechanics the measurement of a quantity interferes with the state of the system. If we measure exactly the position of the particle, then we lose all knowledge of its momentum and vice versa. You should contrast this with the situation in classical mechanics, where we can assume that measurements do not perturb the state of the system.

Summary

- The basic concept introduced in this chapter is that observables are the result of experiments. While this may sound trivial, it is a very important statement. The only knowledge we have of physical systems is the outcomes of experiments. (3.1)
- In quantum mechanics, observables are associated with Hermitian operators acting on the space of physical states. Every time you want to consider an observable, make sure that you know which operator is associated with it. (3.1.2)
- The measurement of an observable can only yield one of the eigenvalues of the operator associated with the observable, independently of the state of the system. (3.2)

- Properties of the spectrum of Hermitian operators. These are extremely important in order to be able to manipulate operators. (3.3)
- The results of an experiment must be considered as a stochastic variable. The state of the system in which the experiment is performed determines the probability of the possible outcomes of measurements. (3.5)
- Measurements in a quantum-mechanical system change the state of the system. This phenomenon is sometimes called collapse of the wave function. (3.6)
- If the operators associated with two observables commute, then the observables are called compatible. Compatible observables have a common set of eigenvectors. (3.7)
- Some operators have a continuous spectrum, and we have introduced the tools to deal with those. Eigenstates of the continuous spectrum are not normalisable. (3.9)
- Definition of the momentum operator, interpretation of the momentum operator as the generator of translations. Construction of normalisable wave packets. (3.10)
- The uncertainty principle is a mathematical consequence of the canonical commutation relations between position and momentum. It is a property of quantum-mechanical systems that does not depend on the quality of the experimental apparatus. (3.11)

Problems

- 3.1** When acting on the wave function, the position operator, $\hat{X}$, corresponds simply to multiplication by x :

$$\hat{X}\psi(x) = x\psi(x).$$

Use the definition of Hermitian conjugation to show that $\hat{X}$ is Hermitian and hence that the potential energy operator $\hat{V} \equiv V(x)$ is also Hermitian.

- 3.2** Prove the following relations:

$$(\hat{f}^\dagger)^\dagger = \hat{f},$$

$$(\hat{f}\hat{g})^\dagger = \hat{g}^\dagger\hat{f}^\dagger,$$

$$[\hat{f}, \hat{g}\hat{h}] = \hat{g}[\hat{f}, \hat{h}] + [\hat{f}, \hat{g}]\hat{h},$$

$$[\hat{f}\hat{g}, \hat{h}] = \hat{f}[\hat{g}, \hat{h}] + [\hat{f}, \hat{h}]\hat{g}.$$

If $\hat{f}, \hat{g}$ are Hermitian, show that $\hat{f}\hat{g} + \hat{g}\hat{f}$ and $i[\hat{f}, \hat{g}]$ are also Hermitian. Show that for any operator $\hat{A}$:

$$\langle A^\dagger A \rangle \geq 0,$$

for any state.

- 3.3** The observables $\mathcal{A}$ and $\mathcal{B}$ are represented by operators $\hat{A}$ and $\hat{B}$ with eigenfunctions $\{u_i(x)\}$ and $\{v_i(x)\}$, respectively, such that

$$\begin{aligned}v_1(x) &= \{\sqrt{3}u_1(x) + u_2(x)\}/2, \\v_2(x) &= \{u_1(x) - \sqrt{3}u_2(x)\}/2, \\v_n(x) &= u_n(x), \quad n \geq 3.\end{aligned}$$

Verify that these relations are consistent with orthonormality of both bases. A certain system is subjected to three successive measurements:

- (i) a measurement of $\mathcal{A}$;
- (ii) a measurement of $\mathcal{B}$;
- (iii) another measurement of $\mathcal{A}$.

Show that if measurement (i) yields any of the values $A_3, A_4, \dots$ then (iii) gives the same result, but that if (i) yields the value A_1 there is a probability of $\frac{5}{8}$ that (iii) will yield A_1 and a probability of $\frac{3}{8}$ that it will yield A_2 . What may be said about the compatibility of $\mathcal{A}$ and $\mathcal{B}$?

- 3.4** Consider a *two-state system*. We denote the two orthonormal states by $|1\rangle$ and $|2\rangle$. In the general case, the Hamiltonian of the system can be written as a 2×2 matrix, where the elements of the matrix are given by

$$H_{ij} = \langle i | \hat{H} | j \rangle.$$

Let us consider the Hamiltonian

$$\hat{H} = \begin{pmatrix} E_0 & -\eta \\ -\eta & E_0 \end{pmatrix}, \quad E_0 \text{ real.}$$

Write the action of $\hat{H}$ on the states $|1\rangle$ and $|2\rangle$.

Show that η has to be real.

Compute the eigenvalues of $\hat{H}$, and the *normalised* eigenvectors.

- 3.5** A two-state quantum system describes a *qubit* in quantum computing. We shall assume that we have specified a basis in the space of physical states, and consider a qubit whose Hamiltonian is described by the matrix

$$\hat{H} = E_0 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Two observables $\mathcal{A}$ and $\mathcal{B}$ are associated respectively with the Hermitian operators A and B that are represented by the matrices

$$\hat{A} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{B} = \begin{pmatrix} 2 & -\sqrt{2}i \\ \sqrt{2}i & 1 \end{pmatrix}.$$

Find the eigenvalues and eigenvectors for $\hat{A}$ and $\hat{B}$.

Are $\hat{A}$ and $\hat{B}$ compatible? Do they commute with the Hamiltonian?

Suppose that an observation of $\hat{A}$ has resulted in $A = 1$, what would be the results for $\hat{B}$, and what would be the respective probabilities?

What would be the probability of finding $A = 1$ if a second measurement is made immediately after the first one?

What is the probability of finding $A = 1$ if a measurement of A is made immediately *after* a measurement of B that yielded the larger eigenvalue of B ?

- 3.6** Let us consider three operators $\hat{f}$, $\hat{g}$ and $\hat{h}$. Show that

$$[\hat{f}, [\hat{g}, \hat{h}]] + [\hat{g}, [\hat{h}, \hat{f}]] + [\hat{h}, [\hat{f}, \hat{g}]] = 0.$$

This result is known as the *Jacobi identity*.

- 3.7** Given the wave function

$$\psi(x) = \left(\frac{\pi}{\alpha^2}\right)^{-1/4} \exp\left(\frac{-\alpha^2 x^2}{2}\right),$$

calculate $\langle x^n \rangle$ and $\Delta x \equiv \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$.

Now calculate the momentum space wave function associated with $\psi(x)$:

$$\tilde{\psi}(p) = \int \frac{dx}{\sqrt{2\pi\hbar}} e^{ipx/\hbar} \psi(x).$$

Using $\tilde{\psi}(p)$, calculate $\langle p^n \rangle$ and $\Delta p \equiv \sqrt{\langle p^2 \rangle - \langle p \rangle^2}$.

With the above results, what do you find for $\Delta x \Delta p$?

- 3.8** Let us consider two Hermitian operators $\hat{A}$, $\hat{B}$, such that

$$[\hat{A}, \hat{B}] = 0,$$

and let us denote by $|\psi_k\rangle$ the eigenfunctions of $\hat{B}$:

$$\hat{B}|\psi_k\rangle = B_k|\psi_k\rangle.$$

Show that $\hat{A}|\psi_k\rangle$ is an eigenstate of $\hat{B}$ with eigenvalue B_k .

In general, there will be a finite number of eigenstates corresponding to the same eigenvalue B_k . We denote these orthonormal eigenstates by

$$\{|\psi_k^{(n)}\rangle, n = 1, 2, \dots, g_k\}.$$

Using the result in (a), deduce that

$$\hat{A}|\psi_k^{(n)}\rangle = \sum_{m=1}^{g_k} A_{nm} |\psi_k^{(m)}\rangle,$$

where A_{nm} are complex numbers.

Show that $A_{nm} = A_{mn}^*$, i.e. that A is a $g_k \times g_k$ Hermitian matrix.

Any finite-dimensional Hermitian matrix can be diagonalised by a unitary transformation U :

$$U^\dagger A U = \begin{pmatrix} A_1 & 0 & \dots & 0 \\ 0 & A_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & A_{g_k} \end{pmatrix}.$$

We can choose the following set of orthonormal linear combinations of $|\psi_k^{(n)}\rangle$, for a given k : $|\phi_k^{(n)}\rangle = \sum_m U_{nm}^\dagger |\psi_k^{(m)}\rangle$.

Show that $\hat{A}|\phi_k^{(n)}\rangle = A_n|\phi_k^{(n)}\rangle$.

We have therefore found a basis of simultaneous eigenstates of $\hat{A}$ and $\hat{B}$:

$$\begin{aligned}\hat{A}|\phi_k^{(n)}\rangle &= A_n|\phi_k^{(n)}\rangle, \\ \hat{B}|\phi_k^{(n)}\rangle &= B_k|\phi_k^{(n)}\rangle.\end{aligned}$$

By repeating the same argument for all eigenvalues B_k , we can explicitly construct a basis of simultaneous eigenvalues of $\hat{A}$ and $\hat{B}$. The two observables are called *compatible*.

- 3.9** The action of the momentum operator $\hat{P}$ on the wave function of a one-dimensional system is given by

$$\hat{P}\psi(x) = \langle x|\hat{P}|\psi\rangle = -i\hbar \frac{d}{dx}\psi(x).$$

Compute

$$\begin{aligned}\hat{P}\hat{X}^n\psi(x), \\ \hat{X}^n\hat{P}\psi(x).\end{aligned}$$

Deduce that

$$[\hat{P}, \hat{X}^n] = -i\hbar n\hat{X}^{n-1}.$$

The operator $\hat{V} = V(\hat{X})$ is defined via the Taylor expansion of the function V :

$$\hat{V} = V(\hat{X}) = \sum_k \frac{1}{k!} V^{(k)}(0) \hat{X}^k,$$

where $V^{(k)}(0)$ are the coefficients of the expansion, i.e. they are numbers computed by evaluating the k th derivative of the function V at $x = 0$. Show that

$$[\hat{P}, \hat{V}] = -i\hbar \frac{d}{dx} V(\hat{X}).$$

- 3.10** Denoting by $|p\rangle$ the eigenstates of momentum with eigenvalue p , the momentum space wave function can be defined as

$$\begin{aligned}\tilde{\psi}(p) &= \langle p|\psi\rangle \\ &= \int dx \langle p|x\rangle \langle x|\psi\rangle \\ &= \int dx e^{-ipx/\hbar} \psi(x).\end{aligned}$$

A generic operator $\hat{O}$ acts on $\psi(p)$ according to

$$\hat{O}\tilde{\psi}(p) = \int dx e^{-ipx/\hbar} \hat{O}\psi(x).$$

Find the action of the momentum operator $\hat{P}$, and the position operator $\hat{X}$ on $\tilde{\psi}(p)$. Check that this new representation of $\hat{X}$ and $\hat{P}$ satisfies the canonical commutation relation.

- 3.11** *Hellmann–Feynman theorem.* Let us consider a system where the potential depends on an external parameter g , $\hat{V} = V(\hat{X}, g)$. Show that for the energy eigenvalues we have

$$\frac{\partial E_n}{\partial g} = \langle \psi_n | \frac{\partial V}{\partial g} | \psi_n \rangle.$$

Remember that the eigenfunctions also depend on g !

This result, known as the Hellmann–Feynman theorem, will be discussed in detail later in the book, in [Chapter 17](#).

- 3.12** *Generalised uncertainty principle.* If ΔA and ΔB denote the uncertainties in the observables $\mathcal{A}$ and $\mathcal{B}$, respectively, in the state $\Psi(x, t)$, then the generalised uncertainty relation states that

$$\Delta A \Delta B \geq \frac{1}{2} ||\langle [\hat{A}, \hat{B}] \rangle||.$$

In order to prove this relation, consider the operators

$$\begin{aligned}\hat{X} &= \hat{A} - \langle \hat{A} \rangle, \\ \hat{Y} &= \hat{B} - \langle \hat{B} \rangle.\end{aligned}$$

The uncertainties in the observables are given by

$$\begin{aligned}(\Delta A)^2 &= \langle \hat{X}^2 \rangle, \\ (\Delta B)^2 &= \langle \hat{Y}^2 \rangle.\end{aligned}$$

For any real number λ , we can construct the state

$$|\phi\rangle = \hat{X}|\psi\rangle + i\lambda\hat{Y}|\psi\rangle.$$

The norm of the state $|\phi\rangle$ is positive by definition:

$$||\phi||^2 = \langle \phi | \phi \rangle.$$

Use this fact to prove the generalised uncertainty relation.